

- 13** Two satellites, A and B, orbiting around Earth have the same kinetic energy. Satellite A has a larger mass than satellite B. Which of the following statements is true?
- A** Satellite A has a larger period.
 - B** Satellite A has a smaller total energy.
 - C** Satellite A has a smaller orbital radius.
 - D** Satellite A has a larger angular velocity.

Ans: A

Since $KE = \frac{1}{2}mv^2$ and $M_A > M_B$, then $v_A < v_B$.

Gravitational force on satellite provides centripetal force for satellite to orbit

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v^2 = \frac{GM}{r}$$

Since $v_A < v_B$, then $r_A > r_B$. Hence C is false.

Since $\omega = \frac{v}{r}$, $v_A < v_B$ and $r_A > r_B$, then $\omega_A < \omega_B$. Hence D is false.

Since $\omega = \frac{2\pi}{T}$ and $\omega_A < \omega_B$, $T_A > T_B$. Hence A is true.

From $v^2 = \frac{GM}{R}$, $KE = \frac{1}{2}mv^2 = \frac{GMm}{2R} = -\frac{1}{2}\left(-\frac{GMm}{R}\right) = -\frac{1}{2}GPE$.

Since $KE_A = KE_B$, then $GPE_A = GPE_B$ and $TE_A = TE_B$. Hence B is false.